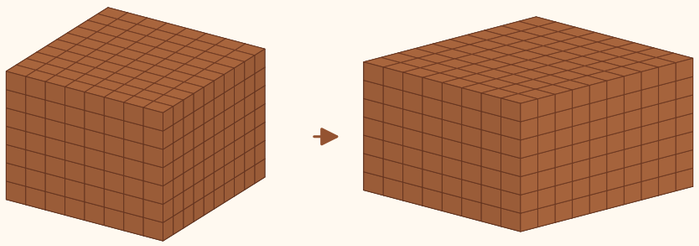
# Ericksen's problem

Determine all universal deformations of homogeneous incompressible isotropic hyperelastic solids in three dimensions.
For every isotropic strain energy, the deformation can be maintained in equilibrium without body forces;
the pressure and surface traction are allowed to depend on the constitutive law.

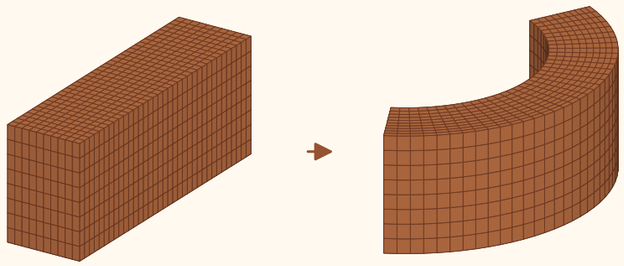
## Universal deformation

$\mathcal{F}_0$ · Homogeneous deformation



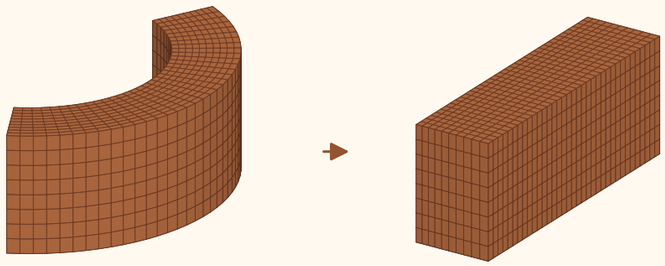
Reference Current

$\mathcal{F}_1$ · Bending of a rectangular block



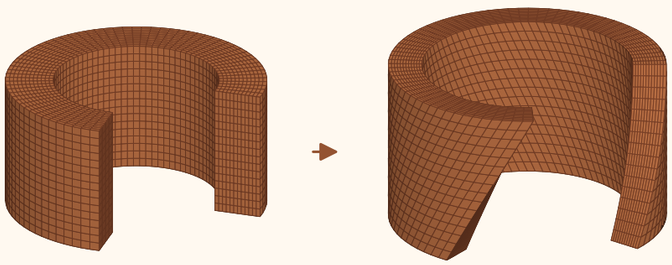
Reference Current

$\mathcal{F}_2$ · Straightening of a cylindrical sector



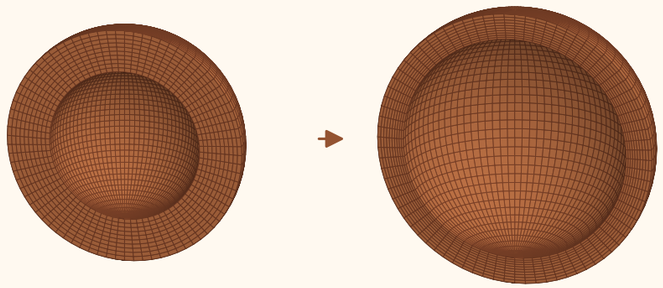
Reference Current

$\mathcal{F}_3$ · Inflation and torsion



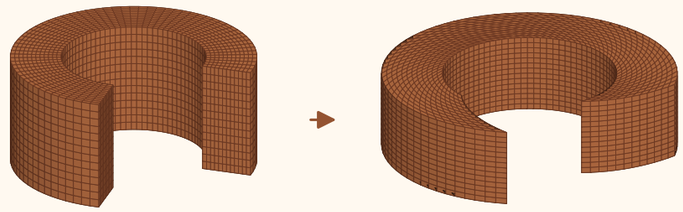
Reference Current

$\mathcal{F}_4$ · Spherical shell deformation



Reference Current

$\mathcal{F}_5$ · Logarithmic cylindrical deformation



Reference Current

### Six classical families · $\det D\chi = 1$

Reference and current configurations use the same physical scale within each panel.

Spherical shells are shown in cutaway.